

STRVX Runway Inspection: Product Breakdown

STRVX Robotics

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Overview

STRVX Runway Inspection is an AI-assisted tool for airport operations teams. A drone or operator captures runway imagery, computer vision flags possible issues, an inspector reviews each one, and approved findings become maintenance tickets. A human stays in the loop at every decision.

Core problem. Ops staff walk the runway before flights, spot issues by eye, log them by hand, write work orders, wait for repair, and reinspect. It is slow, labor-intensive, and produces inconsistent documentation.

MVP goal. A dashboard that runs or reviews drone-based runway inspections, detects four issue categories, presents reviewable candidates per runway and zone, and turns approved findings into maintenance-ready tickets.

Key principle. The model is an assistant, not the authority. The MVP proves STRVX helps teams inspect faster and document better, not that it replaces inspectors. No ticket is finalized on a confidence score alone.

At a glance:

Dimension	Count
Issue categories	4
User roles	3
App screens	10
Services	3
Build phases	5

Primary Users

Three roles, each with a distinct job and landing screen.

Role	Lands on	What they do
Admin	/admin	Create airport, runways, zones, and the map; configure inspection schedules; manage users and roles; set app and stream settings.
Inspector	/	Review findings by runway; approve, reject, or send to manual review; edit the AI ticket draft and severity; create tickets and close after repair.
Maintenance	/ticket	See assigned tickets with images and location; update ticket status; mark repair complete; attach repair notes.

The Workflow

The product replaces a manual walk-down with an AI-assisted review loop.

Today (manual). Ops staff physically inspect each runway, spot issues by eye, log them by hand, and route work orders that are later repaired and reinspected.

With STRVX (AI-assisted). Imagery is captured on a schedule or on demand, CV flags reviewable candidates per runway and zone, the inspector approves a pre-drafted ticket, and the result is exportable as a report.

Inspection-to-closeout pipeline:

1. Schedule or run-now
2. Capture (manual upload or live drone feed)
3. CV plus VLM detection
4. Issue candidates created
5. Human review (approve / reject / manual review / edit)
6. Approved ticket
7. Maintenance repair
8. Inspector closeout
9. Report plus feedback export

Issue Categories

The MVP inspects for four issue types. Each maps to a detection approach.

Category	Examples	Detection approach
Debris / FOD (<code>fod</code>)	Trash, tools, metal, rubber chunks, loose objects	Object / anomaly detection
Pavement damage (<code>pave-ment</code>)	Cracks, potholes, spalling, standing water	Detection / segmentation
Marking issues (<code>marking</code>)	Faded centerline, worn thresholds, rubber buildup	Visual degradation / VLM
Lighting & signage (<code>light-ing</code>)	Damaged or missing lights, obstructed signage	Asset check / VLM

Feature Inventory

Twenty-six tracked capabilities across the product surface. Most of the MVP workflow is shipped on real Postgres; detection quality and external integrations are the open frontier.

Status summary: **19 Built, 4 Partial, 3 Planned.**

- **Built** — shipped on real persistence.
- **Partial** — early or proxy CV models, functional but not airport-trained.
- **Planned** — scheduled for a later build phase.

SETUP & CONFIGURATION

Capability	What it does	Status
Airport / runway / zone setup	Admin defines the physical airfield layout and map	Built
Inspection schedules	Recurring daily window (default 6:00 AM) per airport	Built
Users & roles	Admin / inspector / maintenance accounts (advisory RBAC)	Built
Fleet registry	Drone status, battery, and assignment tracking	Built
App settings	Runtime config such as the live stream URL	Built

CAPTURE & INGESTION

Capability	What it does	Status
Run-now inspection	Create an inspection job on demand, per runway	Built
Manual image upload	Upload runway imagery and tag it to a zone	Built
Live drone feed	HLS video on the Live page with detection overlay	Built
Live capture ingestion	Stream worker files candidates automatically	Built
Drone flight integration	Predefined routes plus auto job creation after a flight	Planned

DETECTION (CV / AI)

Capability	What it does	Status
FOD / debris detection	YOLO object detection for foreign objects	Partial
Pavement damage	Public road-damage detection weights	Partial
Marking degradation	Claude vision (VLM) advisory pass	Partial
Lighting / signage	Claude vision (VLM) advisory pass	Partial
Confidence banding	High / medium / low routing per PRD thresholds	Built
Deterministic fallback	Full workflow still runs with no ML service	Built

REVIEW & TICKETING

Capability	What it does	Status
Issue review	Approve, reject, manual-review, edit category and severity	Built
AI ticket drafting	Suggested ticket text via RL / Claude / template	Built
Ticket lifecycle	Draft to sent to in progress to repaired to closed	Built
Maintenance tracker	Maintenance worklist with repair notes	Built
Reinspection closeout	Inspector closes the ticket after repair (manual)	Built
Automated reinspection	Drone re-fly plus before/after image compare	Planned

REPORTING & LEARNING

Capability	What it does	Status
Inspection report export	Per-inspection JSON / HTML report	Built
Feedback export (JSONL)	Decisions, rejections, and AI-vs-final draft pairs	Built
RL training & eval	Reward model, policy, threshold tuning, eval harness	Built
External work-order integration	Aeros Simple / email / PDF-CSV handoff	Planned

Product Screens

Route	Screen	Audience	Purpose
<code>/</code>	Dashboard / tracker	All	Inspection overview; maintenance sees its worklist
<code>/inspection/[id]</code>	Inspection detail	Inspector	Per-runway status for one run
<code>/runway/[id]</code>	Runway issues	Inspector	Issue candidate list for a runway
<code>/issue/[id]</code>	Issue review	Inspector	Approve / reject / edit a candidate
<code>/ticket/[id]</code>	Work order	Maintenance	Repair details and closeout
<code>/upload</code>	Upload	Inspector / Admin	Manual imagery upload plus zone tag
<code>/live</code>	Live feed	Inspector	HLS feed with live detection overlay
<code>/map</code>	Map	All	Airport / runway map view
<code>/admin</code>	Admin setup	Admin	Airport, runway, zone, schedule, users, settings
<code>/logs</code>	Logs	All	Inspection plus ticket history

Architecture

Three services share one Postgres database.

Service	Port	Stack	Owns
front-end	3000	Next.js App Router + BFF	Browser UI for every screen; proxies most /api/* to the backend; owns uploads, reports, settings, feedback, and live-capture routes
backend	8080	FastAPI + asyncpg	Reads and writes; issues, tickets, airports, drones, runways, zones, schedules, users
ml-service	8000	FastAPI + YOLO + VLM	Upload and live detection; live detection relay (WebSocket); RL draft / reward / threshold plus eval

When the ML service is absent, the app falls back to deterministic detection and ticket drafting so the full workflow still runs.

Data Model & Lifecycles

Core entity chain:

Airport to Runway to Zone to Inspection to Job to Image to Issue candidate to Ticket.

State machines:

- **Inspection:** not_started, in_progress, processing, needs_review, no_issues, tickets_created, completed, failed.
- **Issue candidate:** pending, approved, rejected, manual_review.
- **Ticket:** draft, sent, in_progress, repaired, closed, rejected.

Build Phases

Phase	Goal	Status
Phase 0	Clickable demo with mock data	Done
Phase 1	Image-upload MVP on real Postgres	Done
Phase 2	Drone flight integration	In progress
Phase 3	Work-order integration	Partial
Phase 4	Automated reinspection	Planned

Phase 2 has the live feed and capture path built; predefined routes and auto-flight remain.
Phase 3 has report and feedback export built; the external work-order handoff remains.

Scope Guardrails

Out of scope for the MVP:

- Wildlife detection
- Threat detection
- Voice communications
- Fully autonomous ticketing
- FAA-critical autonomy
- Multi-drone or swarm coordination
- GPS-denied autonomy
- Edge-compute requirement
- Day-one Aeros Simple integration

Direction Read

The product surface and workflow are essentially complete and well-scoped. The remaining value and risk are concentrated in two places: detection quality (the four detectors are functional but proxy-grade and not airport-trained) and the last-mile integrations in Phase 2 and Phase 3 (drone flight automation and the external work-order handoff). The next unit of effort is better spent there than on building more application features.